

Drone-Carried OAI DU Dimensioning Model

OAI CU/DU Split Lab

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Purpose

This document turns the portable-DU idea into an editable sizing model. It does not claim that the airborne deployment has been validated. The lab rollback baseline remains Ethernet CU/DU with SIB8 at OAI commit 102965a669b9444857c27843ec8ce62780bf9d37; the drone work should be treated as a future packaging and flight-integration step after the Quectel F1 path is stable.

The intended architecture is one ground OAI 5GC plus one shared CU, one donor DU for the Quectel modem-serving cell, and one airborne access DU carrying the local USRP access radio. The Quectel modem provides the wireless F1 backhaul to the ground CU. The USRP B210 is the current lab access-radio baseline; the USRP B205mini-i is listed as a lighter future radio candidate, not as a proven drop-in replacement for the current B210 profile.

Editable Formula Model

The sizing model uses component switches, mass assumptions, and power assumptions. For a selected configuration S :

$$m_{\text{fixed}}(S) = \sum_i x_i(S) m_i$$

$$P_{\text{payload}}(S) = \sum_i x_i(S) P_i$$

$$E_{\text{nom}}(S) = \frac{P_{\text{payload}}(S) t_{\text{mission}} r_{\text{reserve}}}{\eta_{\text{dc}} u_{\text{battery}}}$$

$$m_{\text{battery}}(S) = \frac{1000 E_{\text{nom}}(S)}{\rho_{\text{battery}}}$$

$$m_{\text{payload}}(S) = m_{\text{fixed}}(S) + m_{\text{battery}}(S)$$

$$C_{\text{required}}(S) = \frac{m_{\text{payload}}(S)}{\gamma_{\text{payload}}}$$

A drone is considered eligible when $C_{\text{drone}} \geq C_{\text{required}}$. The default $\gamma_{\text{payload}} = 0.70$ keeps the electronics payload below 70.00% of the published drone payload rating, leaving margin for hot weather, altitude, vibration isolation, mounting mistakes, antenna separation, and small later additions.

Table 1: Global sizing assumptions defined at the top of this L^AT_EX file.

Symbol	Current value	Meaning
t_{mission}	20.00 min	Payload electronics run time
r_{reserve}	1.30	Electrical reserve multiplier
u_{battery}	0.80	Usable fraction of nominal pack energy
η_{dc}	0.88	DC/DC conversion efficiency
ρ_{battery}	190.00 Wh/kg	Practical LiPo/Li-ion pack specific energy
γ_{payload}	0.70	Max allowed fraction of drone payload rating

Component Defaults

The values below are deliberately conservative for early design. Vendor-published board weights are used where available. Carrier boards, RF cabling, thermal management, antennas, payload plate, fasteners, and power conversion are grouped as editable engineering allowances because they depend on the mechanical design.

Table 2: Editable component assumptions.

Component	Mass	Power	Source and sizing note
Raspberry Pi 5 board	46.00 g	18.00 W	Board mass should be re-weighed with cooler and storage. Raspberry Pi lists the Pi 5 platform and the official supply is commonly sized around the 5 V, 5 A class. ¹
Jetson Orin Nano carrier class	250.00 g	25.00 W	Editable mass for module, carrier, cooling, and storage. NVIDIA lists Jetson Orin Nano Super Developer Kit power modes from 7 W to 25 W. ²
Mini-PC/x86 class DU	650.00 g	45.00 W	Editable stand-in for the smallest x86 host capable of the chosen OAI real-time profile. ASUS NUC 13 Pro configurations list cTDP classes from 20 W to 45 W depending on CPU. ³
USRP B205mini-i	24.00 g	4.00 W	Light SDR candidate. NI/Ettus B205mini-i published dimensions are 83.3 x 50.8 x 8.4 mm and weight is 24.0 g; it is bus-powered over USB 3.0. ⁴

¹<https://www.raspberrypi.com/products/raspberry-pi-5/>

²<https://www.nvidia.com/en-us/autonomous-machines/embedded-systems/jetson-orin/nano-super-developer-kit/>

³<https://www.asus.com/us/displays-desktops/nucs/nuc-mini-pcs/asus-nuc-13-pro/techspec/>

⁴https://mm.digikey.com/Volume0/opasdata/d220001/medias/docus/822/6002-410-021_Web.pdf

Component	Mass	Power	Source and sizing note
USRP B210	350.00 g	8.00 W	Current lab access-radio baseline. Ettus/Digilent lists 97 x 155 x 15 mm and 350 g for the B210. ⁵
Quectel RM500Q-GL kit	180.00 g	9.00 W	Module plus M.2 carrier, thermal pad/heatsink, SIM holder, USB/PCIe adapter, and antennas. The module itself is about 8.7 g and 30.0 x 52.0 x 2.3 mm. ⁶
Light RF/timing set	180.00 g	6.00 W	Editable allowance for GNSS/timing, filters, short RF cables, antenna mounts, and small fan loads.
B210 RF/timing set	250.00 g	8.00 W	Editable allowance for heavier antenna separation, coax, filters, clocking, and thermal margin.
Power/mount light	300.00 g	included	DC regulators, payload plate, isolation, fasteners, enclosure, wiring, and safety tether.
Power/mount medium	350.00 g	included	Same allowance with stronger B210 vibration isolation.
Power/mount heavy	500.00 g	included	Same allowance for x86 host, larger heatsink, and serviceable field enclosure.

Computed Payload Configurations

Table 3: Computed electronics payload for the default 20.00 minute mission.

ID	Configuration	Fixed mass g	Power W	Battery Wh	Battery g	Payload g
A	Minimum airborne access-DU proof: Pi 5, USRP B205mini-i, Quectel kit	730.00	37.00	22.77	119.86	849.8
B	B210 validation payload: Jetson-class compute, USRP B210, Quectel kit	1 380.00	50.00	30.78	161.97	1 541.9
C	Actual mini-PC class: x86 mini-PC, USRP B210, Quectel kit	1 930.00	72.00	44.32	233.23	2 163.2

⁵<https://digilent.com/shop/ni-ettus-usrp-b210-2x2-70mhz-6ghz-sdr-cognitive-radio/>

⁶https://www.icpdas.com/web/product/download/wireless/modem/3g4g/rm500q-gl/document/data_sheet/RM500Q-GL_datasheet_en.pdf

ID	Configuration	Fixed mass g	Power W	Battery Wh	Battery g	Payload
D	Dual-radio prototype: mini-PC class plus extra donor/access RF margin	2 580.00	100.00	61.55	323.94	2 903.9

Table 4: Drone payload rating required when using the default 70 percent utilization rule.

Configuration	Computed payload kg	Required drone payload rating kg
A	0.85	1.21
B	1.54	2.20
C	2.16	3.09
D	2.90	4.15

Drone Selection Matrix

Payload ratings change by battery choice, temperature, altitude, propellers, and regulatory takeoff weight limits. The values below are planning gates, not purchase approval. Re-check the current manual for the exact region and payload mount before buying or flying.

Table 5: Candidate drones for each configuration.

Drone	Published payload gate	Fits default profiles	Recommendation
Freefly Astro Max	3.0 kg maximum payload; Freefly also lists about 18 min at 3.0 kg payload. ⁷	A: Fit , B: Fit , C: No , D: No	Best compact research drone if the payload is kept near the light or Jetson/B210 profile. Not the first choice for x86 mini-PC packaging.
DJI Matrice 350 RTK	2.7 kg maximum payload and 55 min maximum flight time without payload. ⁹	A: Fit , B: Fit , C: No , D: No	Good mature enterprise platform for the light and B210 validation payloads, especially if the lab wants easier operations before heavy-lift integration.

⁷<https://freeflysystems.com/astro/specs>

⁸<https://freeflysystems.com/knowledge-base/how-long-can-astro-fly>

⁹<https://enterprise.dji.com/matrice-350-rtk>

Drone		Published payload gate	Fits default profiles	Recommendation
DJI Matrice 400		Up to 6 kg payload and 59 min maximum flight time. ¹⁰¹¹	A: Fit , B: Fit , C: Fit , D: Fit	Best first serious choice for the actual x86 mini-PC plus B210 plus Quectel configuration. It keeps enough payload margin for a real enclosure and later RF changes.
Inspired IF1200A	Flight	8.6 kg to 8.7 kg max payload class, with 24.9 kg max gross takeoff weight in current specifications. ¹²¹³	A: Fit , B: Fit , C: Fit , D: Fit	Heavy-lift research choice when the payload needs open integration space, larger batteries, or test instrumentation beyond the radio stack.
Skyfront Perimeter 8 / 8+ hybrid		Skyfront advertises 5 kg payload for 3 h and 10 kg maximum payload for 1 h on the Perimeter 8 family. ¹⁴	A: Fit , B: Fit , C: Fit , D: Fit	Use only when endurance is the primary requirement. It is mechanically and operationally larger than needed for first airborne DU proofs.

Configuration Decisions

Recommended first build

Start with Configuration B if the priority is continuity with the current lab radio baseline. It keeps the USRP B210, uses a smaller compute class than the current mini-PC, and remains eligible for both DJI Matrice 350 RTK and DJI Matrice 400 with the default margin rule. The Matrice 400 is the cleaner first purchase target because it leaves meaningful reserve for enclosure growth, antenna standoffs, RF shielding, heat, and a second modem or timing board.

Minimum-weight research path

Configuration A is the best mechanical path if the USRP B205mini-i proves sufficient for the selected OAI bandwidth and access-cell objective. It reduces the radio board mass from the B210's 350 g class to about 24 g, but it should be treated as an experiment until OAI attach, PWS/SIB8 observation, user-plane traffic, Quectel F1 traffic, and Ethernet rollback are all recorded.

¹⁰<https://enterprise.dji.com/matrice-400>

¹¹<https://www.dji.com/media-center/announcements/dji-release-matrice-400>

¹²<https://www.inspiredflight.com/if1200>

¹³<https://docs.inspiredflight.com/inspired-documentation/products/aircraft/if1200/about-the-if1200/interfaces-and-specs/if1200-specifications>

¹⁴<https://skyfront.com/>

Actual mini-PC path

Configuration C is the closest to carrying the current mini-PC style deployment without pretending the packaging is already optimized. It should skip the Matrice 350/Astro class and go directly to a 6 kg-class or heavier drone. For this configuration, the Matrice 400 is the most balanced option; IF1200A is the reserve-heavy alternative.

Dual-radio or full donor/access prototype

Configuration D models the case where the airborne payload grows beyond one access radio and one modem, for example by adding extra RF, a second SDR path, or field instrumentation. It belongs on Matrice 400 or heavier platforms from the beginning. The IF1200A or Skyfront class becomes attractive only if payload development, endurance, or integration access matters more than portability.

Pre-Flight Engineering Checklist

- Weigh the exact compute board, SDR, Quectel carrier, antennas, coax, heatsinks, payload battery, DC converters, enclosure, and mount before selecting the final drone.
- Recompute this document with the measured masses and with the real OAI power draw under radio load, not idle desktop power.
- Keep the USRP B210 as the verified access-radio baseline until the lighter USRP-mini path has fresh attach, PWS, user-plane, and rollback evidence.
- Keep generated configs, modem logs, packet captures, SIM data, WireGuard keys, and subscriber material out of Git.
- Record sanitized evidence for every airborne trial: OAI commit, drone payload mass, power profile, thermal state, F1-C/F1-U packet proof, Quectel state, phone service/PWS observation, and rollback test.
- Re-check local flight approval, spectrum authorization, payload mounting limits, propeller clearance, and battery transport rules before any outdoor flight.

How to Re-Use This File

Edit the numeric variables near the top of this file, then compile:

```
tectonic docs/drone-du-dimensioning.tex
```

The tables will update automatically. The most important values to change after the first bench packaging pass are `\MissionMinutes`, `\BatterySpecificEnergyWhKg`, `\DronePayloadUtilization`, component masses, and component powers.